

Jinzhe Zhang

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Education



Boston University

09/2023—01/2025

Master's Degree, Major in Computer Science, Data Analytics Concentration

- Core curriculum: Data science with Python, Data Mining, Foundations of Machine learning, Data Visualization

Shanghai Institute of Technology

09/2019—06/2023

Bachelor's Degree (Top 10% GPA)

- Core curriculum: Artificial Intelligence, Python programming, Discrete mathematics, Linear algebra

Skills

- **Languages & Frameworks:** Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Flask), SQL, R, Java
- **Big Data & ETL:** MySQL, PostgreSQL, Spark (Databricks), AWS SageMaker, AWS Lambda
- **Cloud & DevOps:** AWS (SageMaker, Lambda, CloudWatch), Docker, Git
- **ML/MLOps:** Regression, Clustering, LLMs, Model Deployment (APIs), Neural Networks, Deep Learning
- **Tools:** Tableau, Power BI

Professional Experience

Data Scientist Intern | ByteDance (TikTok), Remote | Feb 2023 – Jul 2023

- Optimized video recommendations via **multi-armed bandit experiments**, boosting DAU by **12%** and 7-day retention by **8%**.
- Built **causal inference models** (CUPED, Diff-in-Diff) to evaluate strategy impacts, reducing A/B test bias by **25%** for creator programs.
- Developed **predictive models (XGBoost)** for user Lifetime Value (LTV), achieving a forecast accuracy (**RMSE: 0.18**), directly enhancing revenue predictability.
- Designed **NLP-driven user segmentation** (BERT) for 10M+ creators, increasing top-tier content output by **30%**.
- Created unified metric dashboards (**SQL + Tableau**) for 50+ KPIs, cutting ad-hoc analysis time by **40%**.

Data Scientist Intern | Citibank, Shanghai, China | Aug 2022 – Dec 2022

- Built scalable **Spark/Python ETL pipelines** processing **30K+** financial records daily, enhancing business agility by reducing latency by **35%**.
- Developed **XGBoost/AdaBoost models (GridSearchCV-tuned)** for stock trend prediction, achieving **20% accuracy gain**; results integrated into **Tableau dashboards** for real-time trading decisions.
- Migrated **1TB+** raw data to **AWS S3** with partitioned storage, cutting query latency by **50%** and enabling scalable feature engineering via **Lambda/Step Functions**.
- Collaborated with **quant teams** to align model outputs with **macroeconomic risk frameworks**, driving **15% adoption** of algorithmic trading strategies.

Data Scientist Intern | PDD Holdings Inc., Shanghai, China | May 2022 – Jul 2022

- Developed **logistic regression** and **XGBoost models (Python, Scikit-learn)** to predict patient readmission risks, achieving **88% AUC** and improving generalizability by **15%** via cross-validation.
- Automated **ETL pipelines (Pandas, SQL)** for cleaning **10K+** unstructured EHR records, reducing data prep time by **40%**. Deployed batch inference pipelines via Azure Functions, reducing manual reporting efforts by **22%**.
- Deployed automated Azure-based batch inference pipelines, reducing manual reporting efforts by **22%**.
- Designed **A/B tests (hypothesis testing)** to optimize patient portal UI, cutting churn by **28%** and boosting engagement by **45%**. Built **Power BI dashboards** to visualize risk scores and presented actionable insights.

Project Experience

AWS-Powered Customer Churn Prediction System

10/2024-12/2024

- **End-to-End ML Pipeline:** Built churn models (**GBM, 85% accuracy**) using PySpark for feature engineering on **50GB+** structured/unstructured data (user logs, billing, support tickets).
- **Cloud Deployment:** Built **Dockerized microservices** for model deployment on **AWS SageMaker**, achieving **99%** uptime, deployed as a FastAPI microservice on **AWS SageMaker**, and integrated real-time monitoring via CloudWatch. Reduced churn by **15%** (**\$1M+ revenue retained**).
- **ETL Automation:** Orchestrated data ingestion using Apache NiFi and stored processed data in PostgreSQL for low-latency analytics.